THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 14: **Probability** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Two fair dice are thrown together and the numbers on the upper faces are noted. The probability that the sum of the numbers is 8 is: 1

(a) $\frac{5}{36}$ **(b)** $\frac{1}{6}$ **(c)** $\frac{31}{36}$ **(d)** $\frac{1}{9}$

2. Two fair dice are thrown together and the numbers on the upper faces are noted. The probability that the sum is less than 5 is: 1

(a) $\frac{5}{6}$ **(b)** $\frac{5}{36}$ **(c)** $\frac{7}{36}$ **(d)** $\frac{1}{6}$

3. A bag contains 15 red balls and x blue balls. The probability of drawing a blue ball is 3 times the probability of drawing a red ball. The value of x is: 1

(a) 3 **(b)** 60 **(c)** 44 **(d)** 45

4. A bag contains 7 red and 9 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{4}{7}$. The value of x is: 1

(a) 4 **(b)** 10 **(c)** 5 **(d)** 6

5. A bag contains 11 red and 5 blue balls. 9 more red balls are put into the bag and a ball is drawn at random. The probability that the ball drawn is red is: 1

(a) $\frac{21}{25}$ **(b)** $\frac{19}{25}$ **(c)** $\frac{1}{5}$ **(d)** $\frac{4}{5}$

6. A box contains 70 tickets numbered 1 to 70. One ticket is drawn at random. The probability that the number drawn is a prime number is: 1

(a) $\frac{19}{70}$ **(b)** $\frac{9}{35}$ **(c)** $\frac{2}{7}$ **(d)** $\frac{51}{70}$

7. A lot of 250 ball pens contains 35 defective pens. One pen is drawn at random from the lot. The probability that the pen is good (not defective) is: 1

(a) $\frac{108}{125}$ **(b)** $\frac{107}{125}$ **(c)** $\frac{7}{50}$ **(d)** $\frac{43}{50}$

8. Two fair dice are thrown together and the numbers on the upper faces are noted. The probability that the sum of the numbers is 7 is: 1

(a) $\frac{5}{36}$ **(b)** $\frac{1}{6}$ **(c)** $\frac{7}{36}$ **(d)** $\frac{5}{6}$

9. **Assertion (A):** The probability of getting a number greater than 6 when a die is thrown once is 0. 1
Reason (R): The probability of an impossible event is 0.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The probability of getting a prime number when a die is thrown once is $\frac{1}{2}$. 1

Reason (R): The probability of an event can be greater than 1.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Two fair dice are thrown together and the numbers on the upper faces are noted. Find the probability that the sum is a multiple of 4. 2

12. A bag contains 18 red balls and x blue balls. The probability of drawing a blue ball is 3 times the probability of drawing a red ball. Find the value of x. 2

13. A bag contains 15 red and 9 blue balls. 6 more red balls are put into the bag and a ball is drawn at random. Find the probability that the ball drawn is red. 2

14. Two fair dice are thrown together and the numbers on the upper faces are noted. Find the probability that the first number is greater than the second. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Two fair dice are thrown together and the numbers on the upper faces are noted. Find the probability that the sum is 9. 3

16. A bag contains 9 red and 7 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{2}{3}$. Find the value of x. 3

17. A box contains 84 tickets numbered 1 to 84. One ticket is drawn at random. Find the probability that the number drawn is an even prime number. 3

OR

Two fair dice are thrown together and the numbers on the upper faces are noted. Find the probability that both numbers are even. 3

18. A bag contains 13 red and 11 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{16}{27}$. Find the value of x. 3

OR

A box contains 66 tickets numbered 1 to 66. One ticket is drawn at random. Find the probability that the number drawn is a composite number. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A bag contains 4 red, 5 blue and 6 green balls. One ball is drawn at random. 5
(i) How many equally likely outcomes are there?
(ii) Find the probability that the ball is red.
(iii) Find the probability that the ball is not green.
(iv) Find the probability that the ball is blue or green. If 5 more red balls are added to the bag, what is the new probability of drawing a red ball?

20. Two dice are thrown together. Find the probabilities of these events (in this order): 5
(i) the numbers differ by 2,
(ii) both numbers are even,
(iii) the product is less than 8.

OR

A bag contains 9 red and 21 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{8}{15}$. Find the value of x. 5

END OF QUESTION PAPER — Total: 40 Marks